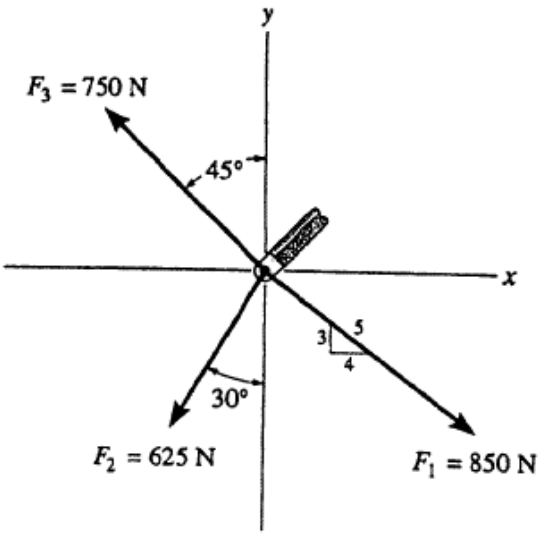
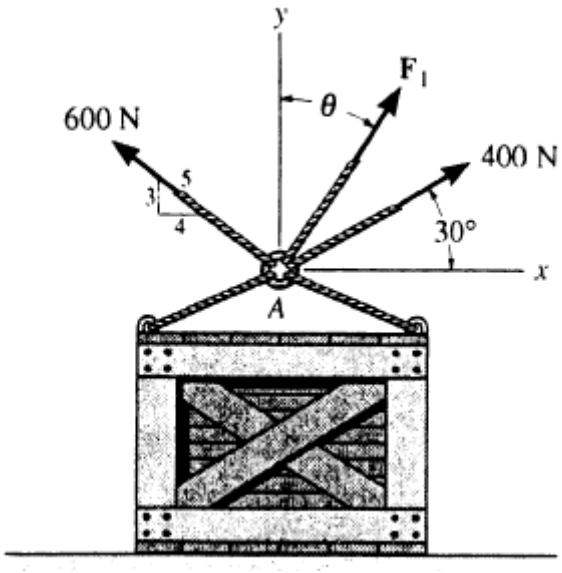
Problems on Chapter (1)

Vector of Mechanics

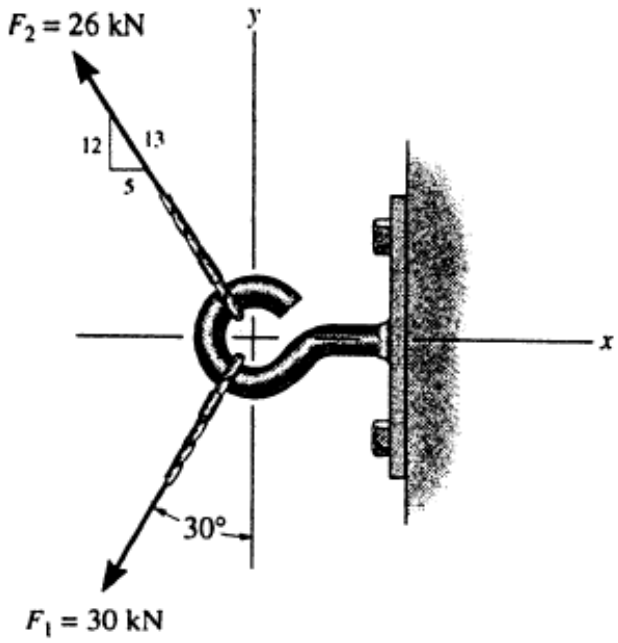
1. Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive x -axis.



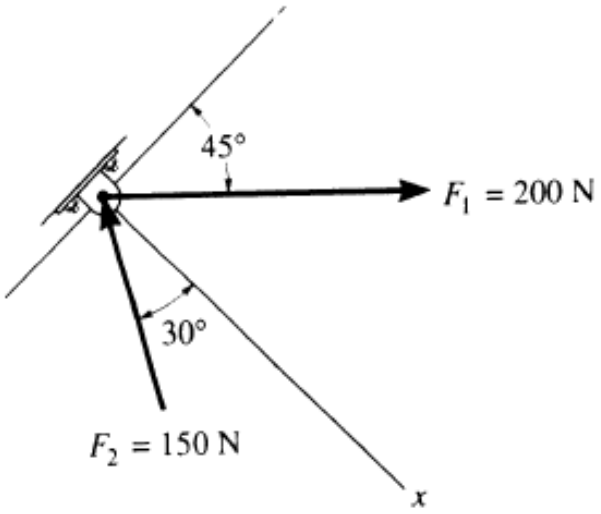
2. Determine the magnitude and direction of F_1, so that the resultant force id directed vertically upward and has a magnitude of 800 N .



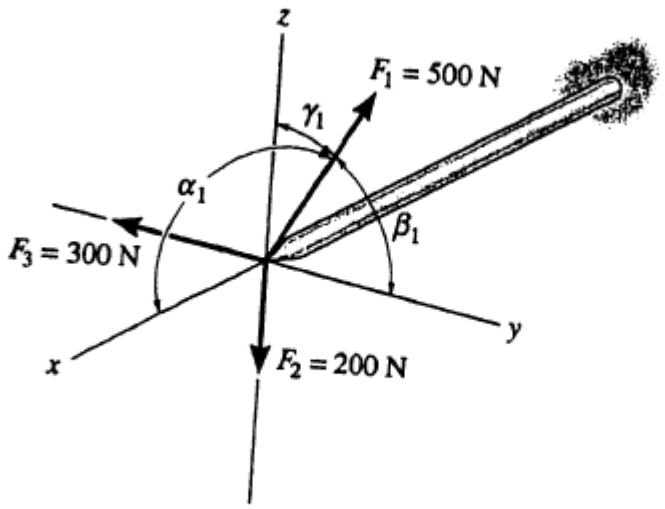
3. Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive x -axis.



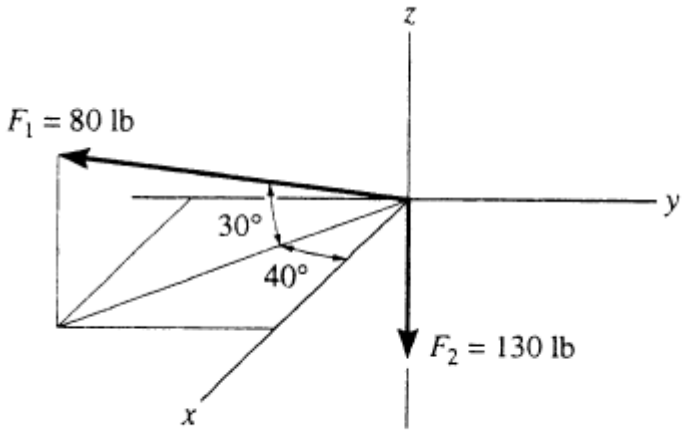
4. Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive x -axis.



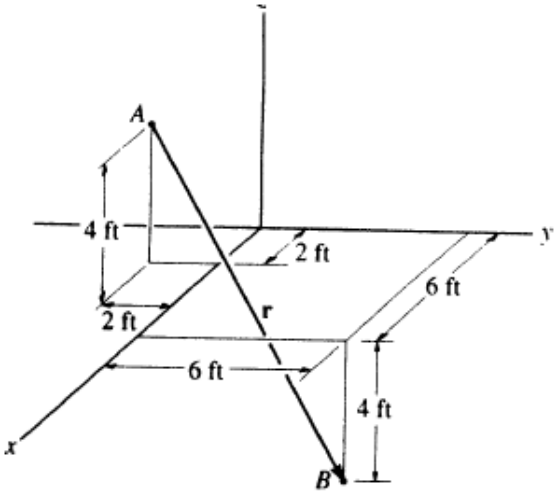
5. Determine the coordinate direction angles α_1, β_1 , γ_1 of F_1 so that the resultant force acting on the mast is F_R = 350 i Newton.



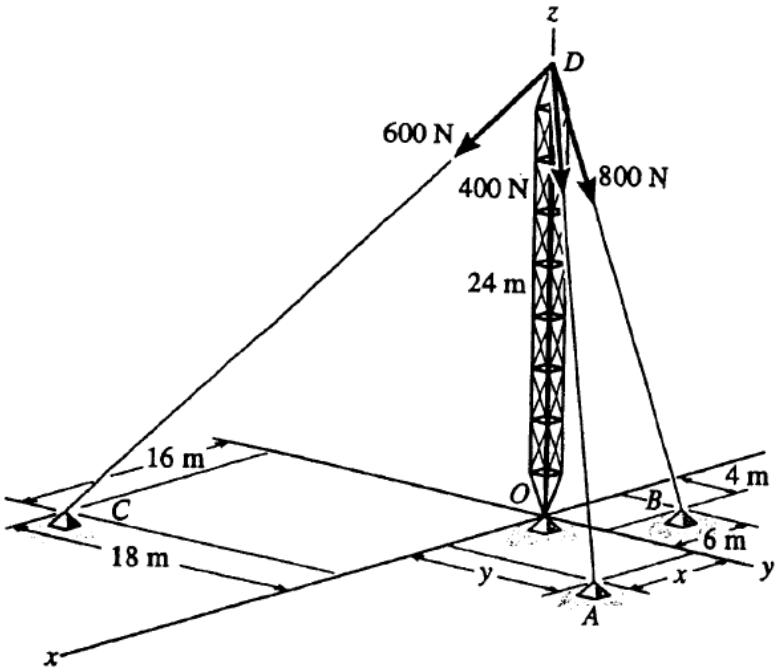
6. Specify the direction angles of F_1 and F_2 and express each force as a Cartesian vector.



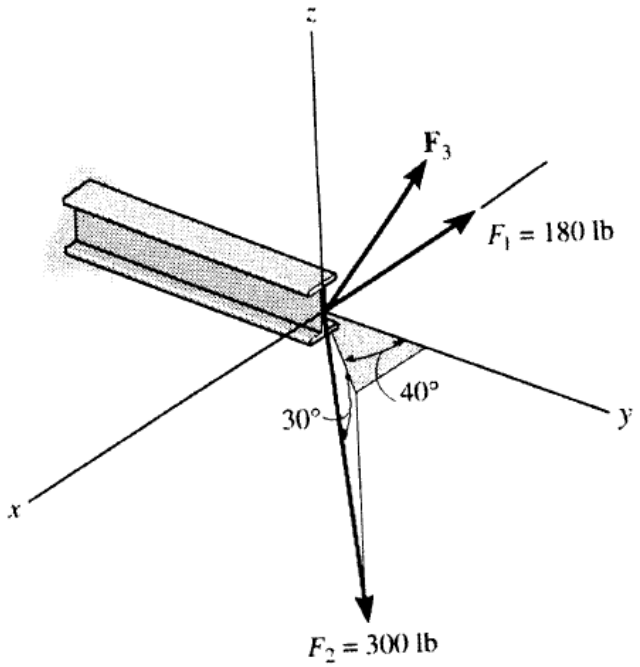
7. Express the position vector r in Cartesian vector form, then determine its magnitude and coordinate direction angles.



8. The tower is held in place by three cables. If the force of each cable on the tower is shown, determine the magnitude and coordinate direction angles α, β , ɣ of the resultant force. Take x= 20 m , y= 15 m.



9. Determine the magnitude and the direction angles of F_3, so that the resultant of the three forces is zero.



10. Determine the magnitude and direction angles of the resultant force acting on A

